

## TRANSMISSION FROM THE BRIDGE

"CIC, report! Lieutenant Ani, are you there?"

"We aren't sure yet, Commander Jaina, we need more time."

"Time is the one thing I cannot give you, lieutenant."

"What does Captain Jacen think about--"

"The captain was hit in the last attack. Helm and navigation are wounded too. They are bypassing our shield harmonics too fast. I need a solution, now!"

"The shields are calibrated by the ship's main battle computer. It's a binary system, right?"

"Yes, so?"

"What if we ran the calibrations through a ternary--"

"INCOMING FIRE!"



## TRANSMISSION ENDS

### Input

You will receive 3 lines of numbers in the format of M.nn, one per line representing the X, Y and Z axis of the ship's shields in base-10 (decimal). M can range from 1 to 100000. All numbers will be positive.

You will also receive a 4th line (note, we are showing it below on several lines, because we don't have as much space on paper as we do in a data file) which has been ternary (base-3) encoded as a baseline check.

In the ternary encoding, letters and connected punctuation are separated by a period and words by a space.

```
123.45
67.89
276.54
2102.10220.11020 10200.11010.11002.11022.11020.11010.11000
11101.10202.10200.11022.11010.11020 10220.11021 11112.10202.11020.11010
11002.10220.11002.10202.11020.1201
```

### Output

For lines 1-3: convert all of the input characters to ternary (base-3) encoded ASCII.

Separate each encoded character with a period.

For line 4, decode the ternary encoded ASCII and output the baseline check in standard ASCII.

```
1211.1212.1220.1201.1221.1222
2000.2001.1201.2002.2010
1212.2001.2000.1201.1222.1221
Air control vector is zero niner.
```

## Discussion

An ASCII to decimal (base-10) table is included in your files. It is named ASCII.txt

To encode an ASCII character into ternary, first look up its value in the table (or use your programming language's ordinal function, if available), then convert that decimal value to ternary.

For instance, "A" is  $65_{10}$  in decimal. 65 in ternary is 2102. "B" is  $66_{10}$ , and 2110 in ternary. "C" is  $67_{10}$ , and 2111 in ternary.

So, "ABC" encoded into ternary, with a period between each character, would be: 2102.2110.2111

Decoding is the same thing, in reverse.  $2102_3 == 65_{10}$ ,  $2110_3 == 66_{10}$ ,  $2111_3 == 67_{10}$   $65 == "A"$ ,  $66 == "B"$ ,  $67 == "C"$ . So, 2102.2110.2111 == "ABC"

Reminder: have you run your solution against **all** of the student data sets?

## Additional Examples

Remember for the 4th line of input, we are showing it below on several lines, because we don't have as much space on paper as we do in a data file.

Input 2

```
9.99
12.12
100000.53
2102.10122.10200.10201.10202.10210.10211.10212.10220.10221.10222.11000.11001.11002.1101
0.11011.11012.11020.11021.11022.11100.11101.11102.11110.11111.11112
```

Output 2

```
2010.1201.2010.2010
1211.1212.1201.1211.1212
1211.1210.1210.1210.1210.1210.1201.1222.1220
Abcdefghijklmnopqrstuvwxyz
```